

Durga Prasad Khatua, PhD

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RESEARCH SUMMARY

Experimental physicist specializing in ultrafast nonlinear optics, femtosecond laser systems, and pulse characterization. Experienced in dispersion management, SHG autocorrelation, and pulse compression for sub-100 fs durations. Strong background in pump-probe spectroscopy and nonlinear exciton dynamics in semiconductors and nanostructures. Skilled in integrating ultrafast optical experiments with vacuum/gas environments, and proficient in numerical modelling of ultrafast pulse propagation and light-matter interactions using Python, MATLAB, COMSOL, Lumerical FDTD, and particle-in-cell simulations (EPOCH). Demonstrated record of peer-reviewed publications, international presentations (CLEO, APS, MRS), and mentoring graduate and undergraduate students.

EDUCATION

PhD, Physics, Homi Bhabha National Institute, India	2022
Thesis Title: "Ultrafast Photoexcited Carrier Dynamics in Two-Dimensional Molybdenum Disulfide."	
M.Sc., Physics, Utkal University, India	2016
B.Sc., Physics Major, Utkal University, India	2013

PROFESSIONAL EMPLOYMENT

Postdoctoral Scholar	Nov 2024 – Present
Physics and Astronomy Department, University of California, Los Angeles, USA	
Postdoctoral Scholar	Oct 2022 – Nov 2024
Mechanical and Aerospace Engineering Department, University of California, Los Angeles, USA	
Graduate Student Researcher	Aug 2017 – Sept 2022
Raja Ramanna Center for Advanced Technology, India	

WORK EXPERIENCE

- **Augmentation of Phase-Sensitive Detection for Low-Repetition-Rate Pulsed Laser Systems | RRCAT**
 - Conceptualized and demonstrated a novel signal detection architecture optimizing ultrafast carrier dynamics tracking in a 1 kHz laser system by seamlessly integrating lock-in amplification with Boxcar averaging techniques.
 - Implemented second-harmonic generation (SHG) autocorrelation spectroscopy for precise pulse characterization and phase-dispersion compensation of femtosecond laser systems.
- **Ultrafast Charge Carrier Dynamics in Semiconductor Nanostructures | RRCAT**
 - Investigated transient carrier dynamics across A, B, and C excitonic transitions in monolayer and multilayer MoS₂ above the Mott density threshold as a function of pump fluence.
 - Elucidated charge transport mechanisms and photoluminescence quenching pathways in Ag nanoparticles, CdTe quantum dots, and engineered Ag-CdTe hybrid nanostructures.
 - Designed, engineered, and aligned multiple custom pump-probe spectroscopy configurations (transmission and reflection geometries) tailored for carrier and spin-relaxation dynamics.

- Developed an integrated optical microscopy system dedicated to high-precision identification of single-layer TMDC flakes and sub-micron beam focusing.
- Characterized morphologic profiles and layer thickness of isolated TMDC flakes using Atomic Force Microscopy (AFM).
- Constructed micro-photoluminescence, micro-absorption, and microscopic transient transmission configurations to probe localized carrier behavior in single-layer TMDCs.
- Contributed to the strategic design and phase-matching alignment of a four-beam coherent optical spectroscopy system.
- **Plasmon-Enhanced High-Energy Light–Matter Interactions for Inertial Confinement Fusion** | UCLA
 - Established a pioneering research initiative investigating high-energy laser interactions with nanoscale and plasmonic metamaterials; verified strong plasmon excitation under ultra-high-intensity regimes utilizing Particle-in-Cell (PIC) simulations.
- **Chip-Scale Cavity Optomechanics for Ultra-Sensitive Thermal Imaging** | UCLA
 - Modeled and optimized optomechanical nanostructures and photonic metamaterials using COMSOL Multiphysics and Ansys Lumerical FDTD to achieve a targeted thermal imaging sensitivity of $80 \text{ fW/Hz}^{1/2}$.
- **Nanosecond Laser–Matter Interaction in the Ablation Regime** | UCLA
 - Engineered and deployed a vacuum-compatible nanosecond (10 Hz) laser ablation system optimized for the study of structured target dynamics.
 - Developed a high-resolution grating spectrometer integrated with an Intensified CCD (ICCD) detector for multi-dimensional plasma imaging and precise spectral diagnostics.
 - Conducted time-resolved emission spectroscopy to chart the spatio-temporal dynamics of plasma plumes during nanomaterial ablation.
- **Millimeter-Wave Doppler Backscattering (DBS) Systems for High-Temperature Plasma Diagnostics** | UCLA
 - Designed and commissioned a V-band 4-channel Doppler Backscattering diagnostic system utilizing independent frequency-sweeping to measure localized turbulence and flow velocity profiles in spherical tokamaks (NSTX).
 - Developed a complementary Q-band 8-channel DBS system to expand radial diagnostic coverage and enable multi-frequency, multi-scale plasma turbulence analysis.
 - Executed rigorous RF/microwave component characterization (including voltage-controlled oscillators, low-noise amplifiers, frequency multipliers, and quadrature mixers) using high-bandwidth oscilloscopes and spectrum analyzers to guarantee high-fidelity signal acquisition.
- **Fast-Ion Instabilities and Ion Cyclotron Emission (ICE) in Tokamak Plasmas** | UCLA
 - Collaborated actively in the MAST-U experimental campaign, contributing to shot planning, real-time data acquisition execution, and cross-diagnostic calibration.
 - Managed on-site diagnostic operations, implementing data quality assurance protocols during active experimental runs.
 - Formulated a novel theoretical framework explaining anomalous ion cyclotron emission (ICE) observed in Ohmic plasmas, fundamentally challenging the long-held paradigm linking ICE exclusively to non-Ohmic energetic ions.

PUBLICATIONS

• Journal Article

1. Khatua, D. P., Singh, A., Gurung, S., and Jayabalan, J., “Pump fluence dependent carrier dynamics at the A-exciton of MoS₂: Monolayer vs. Bulk,” arXiv:2407.05038 (2024).
<https://doi.org/10.48550/arXiv.2407.05038>

2. **Khatua, D. P.**, Singh, A., Gurung, S., and Jayabalan, J., "Excitation density-dependent carrier dynamics in a monolayer MoS₂: Exciton dissociation, formation and bottlenecking," *Micro and Nanostructure* **165**, 207205 (2022). <https://doi.org/10.1016/j.micrna.2022.207205>
 3. **Khatua, D. P.***, Singh, A., Gurung, S., Tanwar, M., Kumar, R., and Jayabalan, J., "A comparative study of ultrafast carrier dynamics near A, B, and C-excitons in a monolayer MoS₂ at high excitation densities," *Opt. Mater.* **126**, 112224 (2022). [* Corresponding author]
<https://doi.org/10.1016/j.optmat.2022.112224>
 4. **Khatua, D. P.**, Singh, A., Gurung, S., Khan, S., Tanwar, M., Kumar, R., and Jayabalan, J., "Ultrafast Carrier Dynamics in a Monolayer MoS₂ at Carrier Densities Well Above Mott Density," *J. Phys.: Condens. Matter* **34** (15), 155401 (2022). <https://doi.org/10.1088/1361-648X/ac4dbf>
 5. Gurung, S., **Khatua, D. P.**, Singh, A., and Jayabalan, J., "Ultrafast carrier dynamics in Ag–CdTe hybrid nanostructure: non-radiative and radiative relaxations," *J. Phys.: Condens. Matter* **33** (18), 185702 (2021). <https://doi.org/10.1088/1361-648X/abeebe>
 6. Gurung, S., Singh, A., **Khatua, D. P.**, Srivastava, H., and Jayabalan, J., "Static and ultrafast optical response of two metal nanoparticles glued with a semiconductor quantum dot," *Photon. Nanostruct. Fundam. Appl.* **43**, 100869 (2021). <https://doi.org/10.1016/j.photonics.2020.100869>
 7. **Khatua, D. P.**, Gurung, S., Singh, A., Khan, S., Sharma, T. K., and Jayabalan, J., "Filtering noise in time and frequency domain for ultrafast pump-probe performed using low repetition rate lasers," *Rev. Sci. Instrum.* **91** (10), 103901 (2020). <https://doi.org/10.1063/5.0010449>
- **Under Preparation**
 1. **Khatua, D. P.**, Shafirin, P., Joly-Jehenne, T., Davoyan, A., "High-Energy Laser-Matter Interaction: Unveiling Plasmon Excitation Dynamics".
 2. **Khatua, D. P.**, Crocker, N. A., T. L. Rhodes, T. L., et al., "Cross-field Extended Coherent Density Structures in the Ion Frequency Range in a Toroidal Magnetized Laboratory Plasma".
 3. Tian, X.*, **Khatua, D. P.***, Tang, Y.*, et al., "ultrafast optical switching of excitons in the multilayer molybdenum disulfide at photo-excitation density above Mott transition". [* Equal contribution]
 - **Conference Proceedings**
 1. **Khatua, D. P.**, Shafirin, P., Joly-jehenne, T., and Davoyan, A., "Ultrafast Plasmon Dynamics Under High-Intensity Laser Excitation," Technical Digest Series (Optica Publishing Group, 2025), paper JPS100_1, CLEO 2025.
 2. Shafirin, P., Feng, P., **Khatua, D. P.**, and Davoyan, A., "Nanophotonic Control of High Energy Light-Materials Interaction," Technical Digest Series (Optica Publishing Group, 2023), paper JTh2A.79, CLEO 2023.
 3. **Khatua, D. P.**, "Ultrafast Photoexcited Carrier Dynamics of Monolayer MoS₂ near Band Edge Above Mott Density," 37th M.P. Young Scientist Congress, March 14–17, 2022.
 4. **Khatua, D. P.**, Gurung, S., Singh, A., Khan, S., and Jayabalan, J., "A comparison of the transient optical response of a monolayer MoS₂ near to its A, B, and C excitons," 65th DAE Solid State Physics Symposium, Bhabha Atomic Research Centre, Mumbai, India, December 15–19, 2021.

CONFERENCE PRESENTATIONS & CONTRIBUTIONS

• Presentations Delivered by Me

1. **Khatua, D. P.**, Crocker, N. A., Micheal, C. A., et al., "Observation of Ion Cyclotron Emission in MAST-U Ohmic Plasma," 67th Annual Meeting of the APS Division of Plasma Physics, Long Beach Convention Center, Long Beach, CA, USA, Nov. 17–21, 2025.

2. **Khatua, D. P.**, Shafirir, P., Joly-Jehenne, T, Davoyan, A., “Ultrafast Plasmon Dynamics Under High-Intensity Laser Excitation,” CLEO 2025, Long Beach Convention Center, Long Beach, CA, USA, May 4-9, 2025.
3. **Khatua, D. P.**, Shafirir, P., Joly-Jehenne, T, Davoyan, A., “Plasmon excitation under extreme-intensity laser pulses,” APS Joint March Meeting and April Meeting, Anaheim, CA, USA, March 16–21, 2025. [Oral Presentation]
4. **Khatua, D. P.**, Gurung, S., Singh, A., and Jayabalan, J., “Ultrafast Carrier Dynamics Near A, B, and C Excitons in Monolayer MoS₂ under High Excitation Densities: A Comparative Study,” APS Joint March Meeting and April Meeting, Anaheim, CA, USA, March 16–21, 2025.
5. **Khatua, D. P.**, Shafirir, P., Davoyan, A., “Plasmon-driven high-energy light-matter interaction,” APS March Meeting, Minneapolis, USA, March 3–8, 2024. [Oral Presentation]
6. **Khatua, D. P.**, Singh, A., Gurung, S., and Jayabalan, J., “Ultrafast Carrier Dynamics at Carrier Densities above Mott Density: Monolayer vs Multilayer MoS₂,” APS March Meeting, Minneapolis, USA, March 3–8, 2024.
7. **Khatua, D. P.**, “Ultrafast Photoexcited Carrier Dynamics of Monolayer MoS₂ near Band Edge Above Mott Density,” 37th M.P. Young Scientist Congress, Indore, India, March 14–17, 2022. [Oral Presentation]
8. **Khatua, D. P.**, Gurung, S., Singh, A., Khan, S., and Jayabalan, J., “A comparison of the transient optical response of a monolayer MoS₂ near to its A, B, and C excitons,” 65th DAE Solid State Physics Symposium, Bhabha Atomic Research Centre, Mumbai, India, December 15–19, 2021. (Best Poster Award)
9. **Khatua, D. P.**, Gurung, S., Singh, A., Khan, S., and Jayabalan, J., “Ultrafast carrier dynamics in a single MoS₂ monolayer flake at high excitation densities,” 7th Theme Meeting on Ultrafast Science, UM-DAE Centre for Excellence in Basic Sciences, Mumbai, India, November 12–14, 2021.
10. **Khatua, D. P.**, Gurung, S., Singh, A., Khan, S., and Jayabalan, J., “Carrier Dynamics Measurement on MoS₂ Monolayers using Ultrafast Pump-Probe Spectroscopy,” Ultrafast Sciences, IIT Bombay, India, November 7–9, 2019.

• Co-authored Conference Contributions

1. **Khatua, D. P.**, Shafirir, P.†, Joly-Jehenne, T, Davoyan, A., “Plasmon Dynamics Under High Intensity Illumination,” MRS Spring Meeting & Exhibit, Seattle, Washington, USA, April 7-11, 2025. [†Presenting Author]
2. Shafirir, P.†, **Khatua, D. P.**, Davoyan, A., “Effects of Thin Films on the Dynamics of Nanosecond Laser Ablation,” MRS Spring Meeting & Exhibit, Seattle, Washington, USA, April 7-11, 2025. [†Presenting Author]
3. Shafirir, P.†, **Khatua, D. P.**, Davoyan, A., “Nanosecond light-matter interaction with structured targets in ablation regime,” APS March Meeting, Minneapolis, USA, March 3–8, 2024. [†Presenting Author]
4. Shafirir, P.†, **Khatua, D. P.**, Feng, P., and Davoyan, A., “Nanophotonic Control of High Energy Light-Materials Interaction,” CLEO: Applications and Technology 2023, San Jose, CA, United States, May 7–12, 2023. [†Presenting Author]
5. Khan, S.†, **Khatua, D.P.**, Khan, S., and Sharma, T. K., “Micro-photoluminescence studies on MoS₂ two-dimensional monolayer using CW and femtosecond laser,” 30th DAE – BRNS National Laser Symposium, Bhabha Atomic Research Centre, Mumbai, India, January 19–22, 2022. [†Presenting Author]

AWARDS AND ACHIEVEMENTS

- 37th MPYSC-Fellowship for Training of Young Scientists (FTYS), M.P. Council of Science and Technology, Govt. of Madhya Pradesh, India 2022

- Best Poster Award, 65th DAE Solid State Physics Symposium (DAE-SSPS) 2021
- Senior Research Fellowship (SRF), Department of Atomic Energy, Govt. of India Research fellowship 2019
- Joint CSIR-UGC test for eligibility for assistance professor in Physics (NET) 2018
- Junior Research Fellowship (JRF), Department of Atomic Energy, Govt. of India Research fellowship 2017
- Graduate Aptitude Test in Engineering (GATE) 2017, 2018
- Joint Entrance Screening Test in Physics (JEST) 2016, 2017
- Post Graduate Merit Scholarship, Govt. of Odisha, Odisha, India 2015, 2016
- Post Graduate Scholarship Scheme in Physical Science, Department of Science and Technology, Govt. of Odisha, India 2015
- Best Graduate of Gopabandhu Science College, Athgarh, India 2013
- Senior Merit Scholarship, Govt. of Odisha, India 2011, 2012, 2013
- National rural talent scholarship, Board of secondary education, Odisha, India 2006

TECHNICAL SKILLS

• Measurement Proficiency

- Transient transmission/reflection spectroscopy and global data analysis PL, and time-resolved)
- Microscopic and confocal pump-probe spectroscopy ◦ Micro-absorbance spectroscopy
- Ultrafast spin dynamics measurements ◦ Raman Spectroscopy
- Photoluminescence (steady-state, micro- ◦ UV-Vis and FTIR spectroscopy
- Atomic force microscopy (AFM)

• Technical expertise

- Femtosecond and nanosecond pulsed laser operation and beam handling
- Vacuum system operation and integration with optical setups
- Design and development of spectroscopic setups for plasma dynamics

◦ Selected Instrumentation & Systems Developed

- * Multi-channel mm-wave Doppler Backscattering systems (V-band 4-channel and Q-band 8-channel) for turbulence and flow diagnostics in spherical tokamaks (NSTX).
- * Ultrafast pump-probe setups (transmission and reflection modes) for carrier and spin dynamics studies in semiconductors and 2D materials.
- * Micro-PL/micro-absorption and microscopic pump-probe systems for transient transmission in single-layer TMDCs.
- * Nanosecond laser ablation system (10 Hz, vacuum-based) with grating spectrometer and ICCD detection for plasma imaging and time-resolved spectroscopy.
- * Optical microscopy system for monolayer TMDC identification and precision beam alignment in pump-probe experiments.
- * Assisted in the development of a four-beam coherent spectroscopy setup.

• Programming and Simulation Knowledge

- **Programming:** Python, MATLAB, Fortran
- **Data Analysis & Visualization:** OriginLab
- **High-Performance Computing:** Parallel computing and HPC workflows
- **Modeling & Simulation:** Particle-in-Cell (EPOCH), Discrete Dipole Scattering (DDSCAT), Computational derivation methods

- **Electromagnetic/Optical Design:** COMSOL Multiphysics (Wave Optics), Ansys Lumerical FDTD, AutoCAD
- **Experimental Facilities and Instruments Operated**
 - **Ultrafast and Nanosecond Lasers:**
 - * Vitara–Legend Elite TOPAS, Coherent (Oscillator–Amplifier–OPA, 35 fs, 1 kHz, 350–1100 nm).
 - * Tsunami, Spectra-Physics (100 fs, 80 MHz, 700–1080 nm).
 - * Mai-Tai HP, Spectra-Physics (85 fs, 80 MHz, 690–1040 nm).
 - * EKSPLA NL300 series nanosecond laser (1064 nm, 3–6 ns, 5–20 Hz).
 - **Vacuum & Sample Environment:** Turbomolecular pump (Agilent).
 - **Microscopy:** Scanning Probe Microscope (Bruker Multimode 8HR).
 - **Electronics & Diagnostics:** Oscilloscopes, Frequency Analyzer (HP 8593E), Lock-in Amplifier (Signal Recovery 7265).

MENTORING

1. Hyeonjong Kim, PhD Scholar, UCLA, USA (Feb–July 2024) – Guided in particle-in-cell (EPOCH) and HPC-based simulations.
2. Joanne Junaed, Undergraduate, UCLA, USA (Aug 2023–Apr 2024) – Mentored in PIC (EPOCH) simulations, HPC workflows, and Python-based data analysis.
3. Swapnasagar Sahu, PhD Scholar, VIT, India (Dec 2022–Dec 2023) – Supervised research on light–matter interaction using DDSCAT.
4. Manish Kumar, Trainee Scientific Officer, RRCAT, India (2020) – Supervised ultrafast auto-correlation experiments.
5. Sonu Kumar, M.Sc. Candidate, RRCAT, India (2019) – Mentored on time-resolved reflectivity with ultrashort pulses.
6. Mohit Raj, M.Sc. Candidate, RRCAT, India (2019) – Guided in transient photoluminescence studies using optical Kerr gate.

REVIEWER ACTIVITY

- *Journal of Materials Science: Materials in Electronics* (8)
- *Semiconductor Science and Technology* (1)
- *Nanotechnology* (4)

MEMBERSHIP

- “American Physical Society (APS) Early Career Membership”: Membership number: 62137939
- “Indian Laser Association Life Member”: Membership number: LM1654

OUTREACH & DIVERSITY ACTIVITIES

- Volunteer at the National Seminar “Ultrafast Science 2018,” Raja Ramanna Centre for Advanced Technology, Indore, India (2018).
- Served as Athletics and Sports Club Secretary at RRCAT, Indore (2018–2020).
- Served as Library Secretary at RRCAT, Indore (2017–2019).
- Volunteer in a UGC-DRS-sponsored national seminar on “Recent trends in physics,” PG Department of Physics, Utkal University, Bhubaneswar, India (2016).
- Volunteer in the “National Science Day Celebration,” PG Department of Physics, Utkal University, Bhubaneswar, India (2016).

- Volunteer in a UGC-DRS sponsored National seminar on "100 YEARS OF GENERAL THEORY OF RELATIVITY," PG Department of Physics, Utkal University, Bhubaneswar, India (2015).
- Volunteer and organizing committee member in a national seminar on "Space exploration and Interdisciplinary Science," Gopabandhu Science College, Athgarh, India (2012).

PROFESSIONAL SERVICE

- **Reviewer Activity:** Journal of Materials Science: Materials in Electronics (8), Semiconductor Science and Technology (1), Nanotechnology (4).
- **Professional Memberships:** American Physical Society (APS) Early Career Member (ID: 62137939); Indian Laser Association Life Member (ID: LM1654).
- **Outreach & Diversity:**
 - Volunteer and organizing roles in multiple national seminars and outreach programs, including: Ultrafast Science 2018 (RRCAT, Indore); UGC-DRS National Seminars on "Recent Trends in Physics" (2016) and "100 Years of General Theory of Relativity" (2015) at Utkal University, Bhubaneswar; National Science Day Celebrations (Utkal University, 2016); "Space Exploration and Interdisciplinary Science" (Gopabandhu Science College, 2012).
 - Leadership roles at RRCAT, Indore: Athletics and Sports Club Secretary (2018–2020) and Library Secretary (2017–2019).